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S001R04.edf
2.5MB



Model



csp_lda



Ground-truth labels

- T1 = imagined **left** fist
- T2 = imagined **right** fist
- T0 = rest (not classified)



EEG Motor Imagery Prediction App

PhysioNet EEG Motor Movement / Imagery Dataset • Left-hand vs Right-hand motor imagery (runs 4, 8, 12)



File Information

Sampling Frequency

160 Hz

Channels

64

Duration

125.0 s

All channel names



EEG Signal Visualisation

Channel

C3

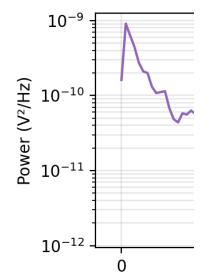
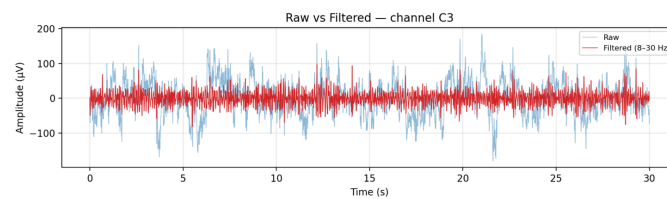
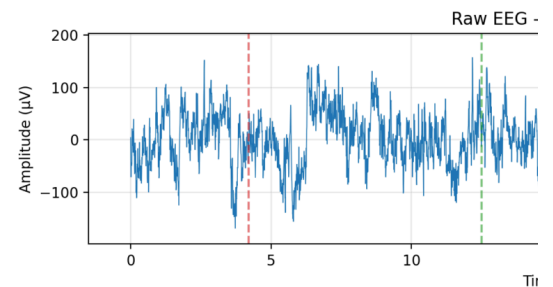


Time range (s)

0.00 30.00

0.00

124.99



Preprocessing, Feature Extraction, and Model Inference

1. Band-pass filter

8–30 Hz (FIR)

2. Extract events

T1 (left) and T2 (right)
from EDF annotations

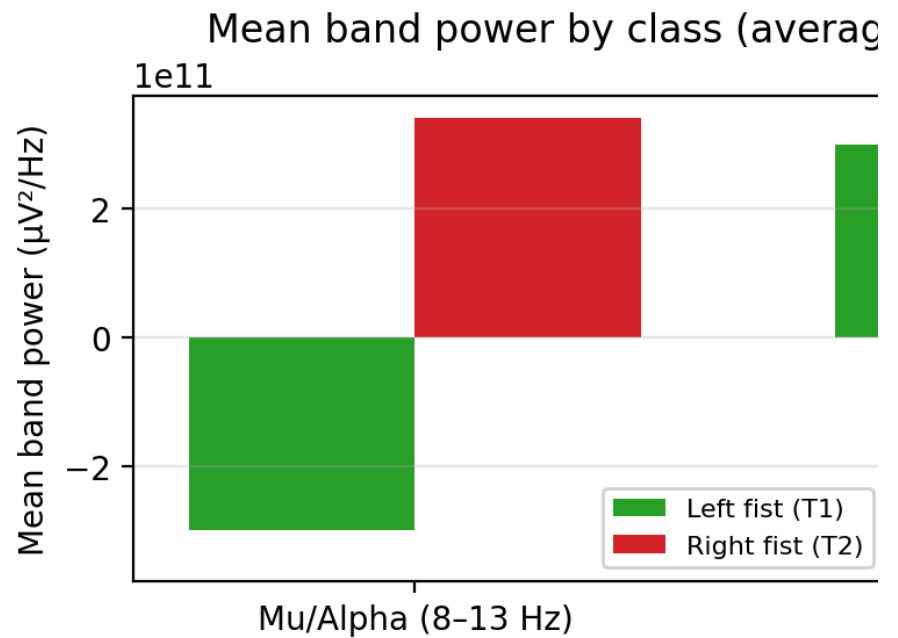
3. Create epochs

Window 0.5 → 2.5 s
around each event

4. Feature extraction

We extract power features from each epoch.

Extracted **15 epochs** (8 left · 7 right) · band-power feature vector = **20** (9 motor-cortex channels × 2 band-pass filters)



Offline Model Training Summary

Training subjects: 40 (S001–S040) · **Runs:** [4, 8, 12] · **Strategy:** Within-subject 80/20 split per subject, performance (standard BCI evaluation). Cross-subject 5-fold CV also reported.

Preprocessing: band-pass 8.0–30.0 Hz · epochs 0.5 → 2.5 s · channels: Band-power: 9 motor-cortex cl 'CP4']; CSP/Riemannian: ALL 64 EEG channels (spatial filtering)

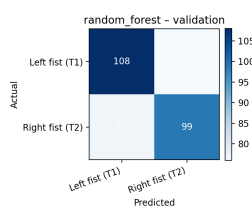
Features: None · bands ['mu_alpha', 'beta'] · 20 features/epoch

Random Forest

Accuracy
0.575

F1-score
0.575

Within-subject eval · cross-subject CV: 0.537 ± 0.022



Mlp

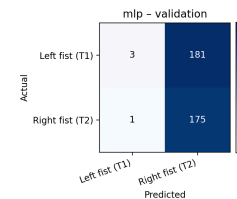
Accuracy

0.494

F1-score

0.338

Within-subject eval · cross-subject CV: 0.561 ± 0.013



Csp Lda

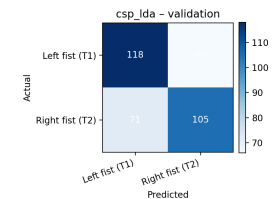
Accuracy

0.619

F1-score

0.619

Within-subject eval · cross-subject CV: 0.593 ± 0.010



Rie

Accuracy
0.5

Within-subject eval · cross-subject CV: 0.537 ± 0.022

Prediction Results and Evaluation

Accuracy (this file)

0.800

F1-score (this file)

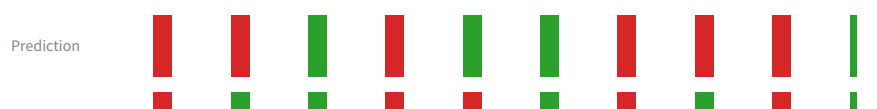
0.800

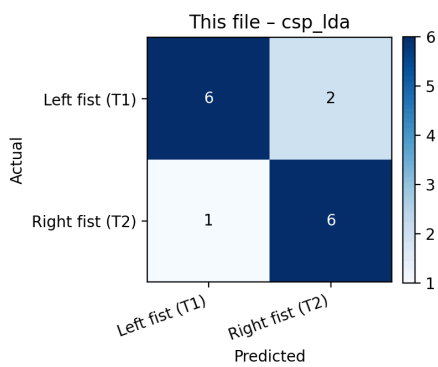
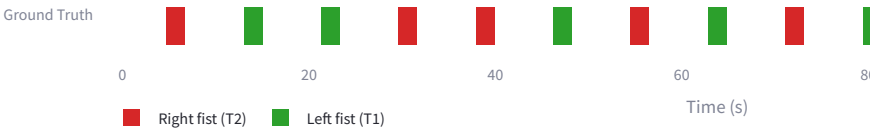
Correct epochs

12 / 15

Ground-truth vs Prediction timeline

Ground Truth vs Prediction Timeline





Epoch	Start (s)	End (s)	C
1	4.7	6.7	F
2	13	15	L
3	21.3	23.3	L
4	29.6	31.6	F
5	37.9	39.9	F
6	46.2	48.2	L
7	54.5	56.5	F
8	62.8	64.8	L
9	71.1	73.1	F
10	79.4	81.4	L

Tip: download the table or compare against the timeline above to see which time windows the model gets right or